



1984 - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational Wave Sources and r-Process Nucleosynthesis

Cycle: 1, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) TOO1
	3		NIRCam Imaging	(1) TOO1
	2		NIRSpec Fixed Slit Spectroscopy	(2) TOO2
	4		NIRSpec Fixed Slit Spectroscopy	(2) TOO2

ABSTRACT

The joint gravitational wave and electromagnetic detections of the binary neutron star (BNS) merger GW170817 ushered in a new era of astrophysics. In the UV/optical/NIR the emission (a "kilonova") was powered by radioactive decay of nuclei produced via r-process nucleosynthesis. In the gamma-ray, X-ray, and radio the emission was instead powered by an off-axis jet typical of short gamma-ray bursts (SGRBs); this connection was previously supported by the HST detection of a kilonova in the short GRB130603B. With only a single joint GW-EM detection and a single kilonova detection in an SGRB, the key frontier is to begin to map the distribution of merger outcomes: ejecta mass, velocity, geometry, and nucleosynthetic yields. Here we propose to achieve this goal with detailed JWST observations of a kilonova associated with an SGRB; observations of kilonovae in SGRBs are essential because the orientation is well known (face-on, along the binary's angular momentum axis) and the LIGO/Virgo Observing Run 3 did not yield any new joint GW-EM detections. Such a detailed study can only be achieved at the redshifts typical of SGRBs with the infrared sensitivity and angular resolution of JWST. We request up to 11.9 hours for 1 SGRB event. The JWST observations will be supported by an on-going multi-wavelength program (e.g., Chandra, VLA, ALMA, Gemini, Keck, Magellan, MMT) that will provide the target, cover optical and early NIR follow-up to establish the baseline behavior, and complete the multi-wavelength picture of the event. Given the broad interest in this topic we waive the proprietary period.

OBSERVING DESCRIPTION

The program will target a single short gamma-ray burst (SGRB) at a redshift of $z < 0.9$. The first epoch will be timed to an observer time of $4x(1+z)$ days, requiring a disruptive ToO at 4-8 days. Depending on the brightness of the host galaxy and the angular separation between the SGRB and its host, we may need to obtain template observations at >1 month (non-disruptive ToO) for host subtraction, with an identical setup and time request to the main observation.

At $z < 0.5$ we will obtain NIRSpec spectroscopy with the following setup: prism grating with the S200 A1 slit, full frame, NRS IRS2 readout pattern,

JWST Proposal 1984 (Created: Thursday, February 17, 2022 at 2:00:47 PM Eastern Standard Time) - Overview

4 integrations per exposure with 25 groups per integration, and 2 dithered exposures (14,706 s on-source; 21,362 s with overheads). This will provide $S/N=5-15$ at 0.85-4 micron, and with expected binning by a factor of 2-3 (due to the broad spectral features) it will achieve $S/N>8$ at 0.85-4 micron. At $z=0.5-0.9$ we will obtain NIRCcam imaging with the following setup: We will pair the short- and long-wavelength filters as F115W+F444W, F150W+F356W, and F200W+F277W. The first pairing is specifically designed since F115W and F444W have lower throughput relative to other filters, so we will use a longer integration time in those filters. We use 5497 sec on-source for F115W+F444W (7 groups, 1 integration, 4 dithered exposure, DEEP8 readout), and 3779 sec on-source for F150W+F356W and F200W+F277W (5 groups, 1 integration, 4 dithered exposure, DEEP8 readout). The resulting limiting magnitudes ($S/N=3$) are 29.2-29.6 AB mag and $S/N=10$ detections will be achieved at 27.9-28.3 AB mag (0.1" radius aperture). The total on-source time is 13,060 s (19,398 s with overheads).

Proposal 1984 - Targets - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational Wave...

Generic Targets	#	Name	Criteria	Description
	(1)	TOO1	SGRB-KN at $0.5 < z < 0.9$	
	(2)	TOO2	SGRB-KN at $z < 0.5$	
	(3)	ACQSTAR	Point source with F115W < 23 mag near the SGRB-KN	

Proposal 1984 - Observation 1 - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational...

Observation	Proposal 1984, Observation 1										Thu Feb 17 19:00:47 GMT 2022
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Observation 1) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted. (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria	Description							
	(1)	TOO1	SGRB-KN at 0.5 < z < 0.9								
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F444W	DEEP8	7	1	4	4	5497.226	60604.1	
	2	F150W	F356W	MEDIUM8	9	1	4	4	3779.343	60604.3	
	3	F200W	F277W	MEDIUM8	9	1	4	4	3779.343	60604.5	
Special Requirements	On Hold pending target coordinates Target Of Opportunity response time 4 Days										
	3 After 1 by 30 Days to <None specified>										

Proposal 1984 - Observation 3 - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational...

Observation	Proposal 1984, Observation 3										Thu Feb 17 19:00:47 GMT 2022
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Comments: Template observation if required to remove host galaxy contamination from Observation 1										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria				Description				
	(1)	TOO1	SGRB-KN at 0.5 < z < 0.9								
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F444W	DEEP8	7	1	4	4	5497.226	60604.1	
	2	F150W	F356W	MEDIUM8	9	1	4	4	3779.343	60604.3	
	3	F200W	F277W	MEDIUM8	9	1	4	4	3779.343	60604.5	
Special Requirements	On Hold template for Observation 1										
	3 After 1 by 30 Days to <None specified>										

Proposal 1984 - Observation 2 - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational...

Observation	Proposal 1984, Observation 2											Thu Feb 17 19:00:47 GMT 2022
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Observation 2) Warning (Form): Response time < 14.0 Days is disruptive to the scheduling process. Only a limited number of disruptive ToOs will be accepted.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria				Description					
	(2)	TOO2	SGRB-KN at z < 0.5									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	3 ACQSTAR	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	60604.8	
Template	Slit				Subarray							
	S200A1				FULL							
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2	25	4	1	NONE	2	8	14705.601	60604.7

Proposal 1984 - Observation 2 - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational...

Special Requirements	On Hold pending target coordinates Target Of Opportunity response time 4 Days 4 After 2 by 30 Days to <None specified>
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Proposal 1984 - Observation 4 - Fine-Tuned Search for Kilonova Emission in a Short Gamma-Ray Burst: Implications for Gravitational...

Observation	Proposal 1984, Observation 4											Thu Feb 17 19:00:47 GMT 2022
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	Comments: Template observation if required to remove host galaxy contamination from Observation 3											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria				Description					
	(2)	TOO2	SGRB-KN at z < 0.5									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	3 ACQSTAR	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	60604.8	
Template	Slit					Subarray						
	S200A1					FULL						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2	25	4	1	NONE	2	8	14705.601	60604.7

Special Requirements	On Hold template for Observation 3 4 After 2 by 30 Days to <None specified>
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